

DEEP RESEARCH REPORT

TradePilot

Next-Gen Indian Trading Platform

Competitor Analysis • Market Gaps • Technology Architecture •
Regulation • Community Insights

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PROJECT INFORMATION

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Executive Summary

The Indian retail trading market is at an inflection point. With **212M+ demat accounts**, 33% YoY growth, and a post-COVID boom in retail participation — the market is exploding. But there's a dark side: **93% of F&O traders lose money**, with aggregate losses exceeding Rs 1.8 trillion over 3 years (SEBI, 2024).

Current platforms (Zerodha, Groww, Upstox) are **order execution terminals** — they help you place trades but don't help you make money. Zerodha sits at 1.7/5 on Trustpilot. Upstox at 1.6/5. A 6+ lakh user exodus happened in mid-2025.

TradePilot's thesis: Build the first Indian trading platform where every screen, every feature asks: *"Will this trade make you money?"* — AI-powered trade scoring, risk guardrails, strategy validation, and profit probability before you click buy.

212M+

DEMAT ACCOUNTS

93%

F&O TRADERS LOSE

Rs 1.8T

AGGREGATE
LOSSES

1.7/5

ZERODHA TRUST
SCORE

1. Market Sizing & Opportunity

1.1 TAM / SAM / SOM

Metric	Value	Source
Demat accounts (2025)	212.8M	CDSL/NSDL
New accounts/month	3.8M	Industry data
Active trading accounts	154M	NSE
Active individual traders	10M+	SEBI
Security brokerage TAM (2025)	\$2.20B	IMARC Group
Brokerage TAM (2034 proj.)	\$2.83B	IMARC Group
Retail AUM (mid-2024)	Rs 61.33L cr	Industry data
F&O active traders	11.3M	SEBI
F&O aggregate losses (FY22-24)	Rs 1.8T	SEBI Study

1.2 Growth Trajectory

- **5-year demat CAGR (2019-2024):** ~32% annually
- **2024 growth:** 33% YoY (46M new accounts)
- **Individual traders:** 120% increase FY22-FY25
- **Registered investors:** Tripled since March 2020

1.3 The Paradox

Account growth is explosive (33% YoY) but brokerage revenue growth is anemic (2.84% CAGR). This means: **high user acquisition, low monetization, high churn risk**. Platforms are commoditized — the winner will be whoever solves profitability for traders, not just access.

1.4 Key Demographics

Segment	Data	Implication
Age	43% of consumption under 25 by 2025	Massive young cohort, needs education
Income	75% of F&O traders earn <Rs 5L/year	Price-sensitive, high-risk behavior
Gender	24.3% female nationally	Emerging underserved segment
Geography	50% new accounts from Tier-II/III	Rural/semi-urban growth outpacing metros
Loss rate	93% of F&O traders lose money	CRITICAL — the problem to solve

2. Competitor Landscape

2.1 Indian Platform Comparison

Platform	Strengths	Weaknesses	Trust
Zerodha Kite	Largest base (2.3cr+), free APIs, ecosystem (Sensibull/Streak/Smallcase)	Limited charting, no native options tools, outages	1.7/5
Groww	Simplest UX, 1.25cr active users, 100% Trustpilot resolution	No commodities, no algo, basic charting, no bracket orders	Better
Upstox	3-in-1 account, NRI support, TradingView integration	Hidden charges, "dead slow" app	1.6/5
Angel One	SmartAPI for algo, commodities, NRI trading	Steep learning curve, highest AMC, less beginner-friendly	Mid
Dhan	Fastest execution (<50ms), strong options focus (DEXT engine)	Smaller platform, limited brand recognition	Growing
Fyers	Best native charting/TA tools, lifetime free AMC	Mid-tier user base, less documented API	Niche
5Paisa	Transparent flat pricing (Rs 20/trade)	Limited charting, smaller ecosystem	Low

2.2 International Benchmarks

Platform	What India Lacks	Feature Gap
Interactive Brokers	47K+ stocks, 160+ markets, 0.52ms latency	Global access, ultra-low latency, margin lending
ThinkorSwim	Advanced charting, backtesting, paper trading with slippage	Strategy simulation, conditional orders
TradingView	1000+ indicators, Pine Script, community scripts	Professional charting, custom indicators
Robinhood	Fractional shares, no minimums	Micro-investing, zero-barrier entry
QuantConnect	100TB+ backtesting data, heatmap optimization	Quantitative strategy validation
Composer	No-code hedge fund strategy builder	Visual strategy building for non-coders
Bookmap	Advanced order flow visualization	DOM heatmaps, liquidity analysis

2.3 The 10 Critical Gaps No Indian Platform Fills

1. **Professional-grade charting & backtesting** — forced to use TradingView separately
2. **Global asset access** — no unified India + international markets
3. **Ultra-low latency** — 50ms vs 0.52ms internationally
4. **Advanced options strategy builder** — must use Sensibull/Opstra separately
5. **Fractional shares / micro-investing** — can't build diversified portfolio with small capital
6. **Integrated backtesting & paper trading** — zero Indian platforms offer this
7. **Unified trading account** — separate accounts for different asset classes
8. **Sophisticated risk management** — no OCO, trailing stops inconsistent, no TWAP/VWAP
9. **Ecosystem integration** — 3-4 tabs/apps for complete workflow
10. **Tax automation** — manual Excel reconciliation for capital gains

3. Community Pain Points

3.1 Top 15 Pain Points (Ranked)

#	Pain Point	Severity	Impact
1	Platform crashes during volatility	Critical	Financial losses from stuck orders
2	Order execution failures & slippage	Critical	Orders stuck in limbo, price lag
3	Poor customer support	High	No resolution, automated replies
4	Hidden charges & unclear fees	High	"Zero brokerage" masks real costs
5	Inadequate charting tools	High	Forced to split between broker + TradingView
6	No backtesting or paper trading	High	Deploying real capital without validation
7	Missing portfolio analytics	Medium	No P&L attribution, drawdown analysis
8	Poor options trading tools	High	No Greeks, no multi-leg from platform
9	No strategy automation for retail	High	Must use 3rd-party (Tradetron, Streak)
10	Slow settlement & withdrawal	Medium	T+3 frustration
11	Zero guardrails for beginners	Medium	No loss limits, no position sizing guidance
12	Mobile app UX problems	Medium	Dead slow performance during trading
13	Tax reporting nightmare	Medium	Manual capital gains tracking
14	No social/community features	Medium	Traders scattered across Telegram/Discord
15	Data accuracy issues	Low	Chart prices differ from execution prices

3.2 The Workaround Stack

Today's profitable trader uses **5-6 separate tools**:

```

Broker (Zerodha/Dhan)    -- order execution
+ TradingView            -- charting (Rs 500+/month)
+ Sensibull/Opstra       -- options Greeks & strategy
+ Tradetron/Streak       -- automation
+ Excel                  -- portfolio tracking & P&L
+ Moneycontrol/Screener  -- fundamental analysis
+ Cleartax               -- tax filing

```

TradePilot's opportunity: Unify this into ONE platform.

3.3 Dream Features Traders Want

1. Integrated backtesting engine (historical options backtesting)
2. Unified portfolio analytics dashboard (P&L by trade/strategy/sector)
3. Built-in strategy automation with risk guardrails
4. TradingView-grade charting natively
5. Tax automation (capital gains categorization, ITR export)
6. AI-powered risk warnings ("You're risking 5% on this trade")
7. Community layer with verified signal providers
8. Advanced order types (OCO, bracket, conditional)
9. Greeks analysis integrated with order flow
10. Multi-broker integration (execute across brokers from one dashboard)

4. Technology Architecture

4.1 Recommended Tech Stack

Frontend:

Backend:

Event Bus:

Time-Series DB:

Analytics DB:

ML/AI Pipeline:

Real-Time:

Broker APIs:

Deployment:

React/Next.js + TradingView Charting Library + WebGL (heatmaps)

Go/Rust microservices + gRPC (risk validation)

Apache Kafka (multi-broker event streaming)

QuestDB (tick data, 12-36x faster than InfluxDB)

TimescaleDB (portfolio analytics, continuous aggregates)

PyTorch (price models) + spaCy (NLP sentiment)

WebSocket + Server-Sent Events (fallback)

Zerodha Kite Connect v3 + Angel SmartAPI + Dhan API

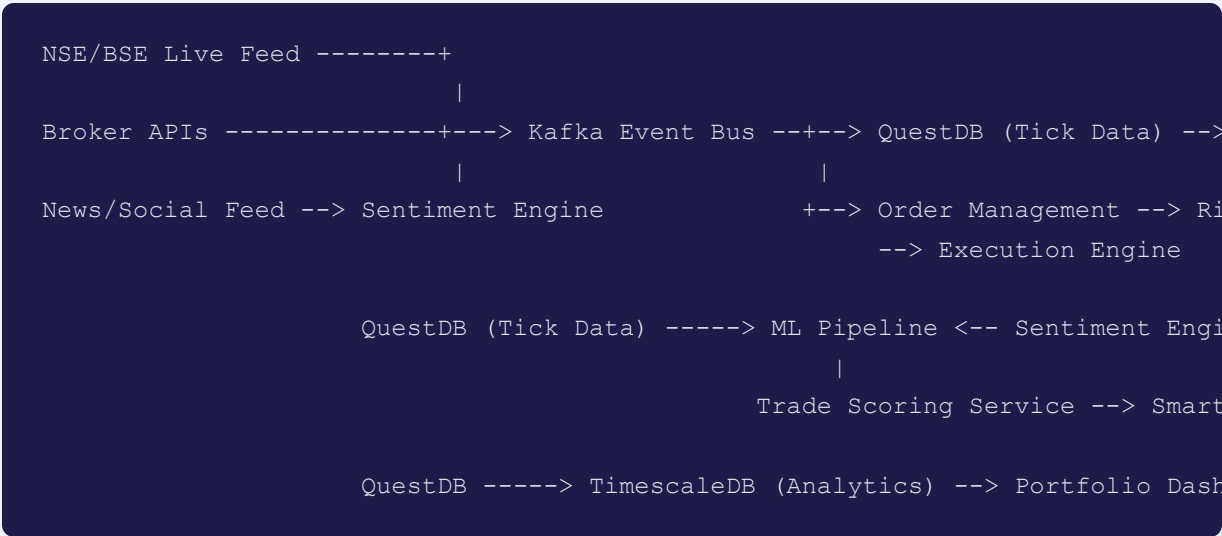
Docker + Kubernetes (AWS Mumbai region)

4.2 AI/ML Capabilities

Capability	Technology	What It Does
Price prediction	LSTM/Transformer models	30-day price movement forecasting
Sentiment analysis	NLP on news + FII/DII flows	Market mood scoring
Pattern recognition	CNN-based chart analysis	Auto-detect patterns with probability scores
Trade risk scoring	Ensemble models	Per-trade probability of profit/loss
Smart alerts	Contextual AI	Alert when sentiment diverges from price
Position sizing	Kelly criterion + volatility	Optimal position size per trade

4.3 Data Architecture

ARCHITECTURE DIAGRAM – DATA FLOW



4.4 Interactive UI Features

- **Liquidity heatmaps** — real-time order flow visualization (Bookmap-style)
- **Options payoff simulator** — interactive Greeks with strategy overlay
- **Risk meter** — portfolio-level heat map with position risk scoring
- **Parameter sensitivity heatmaps** — backtest all strategy iterations visually
- **Flow visualization** — historical footprint charts showing volume at price
- **No-code strategy builder** — Composer-style drag-and-drop

5. Regulatory Roadmap

5.1 Path to Market (3 Options)

Path	Timeline	Cost	Risk	Best For
SEBI Research Analyst (RA)	1-2 months	Rs 5-10L	Low	MVP — algo signals/advisory
Authorized Person (AP)	2-4 weeks/exchange	Rs 2-5L/broker	Medium	White-label under broker
Full Broker License	6-18 months	Rs 50L+	High	Full independence

5.2 Recommended Phased Approach

Phase 1 (Month 0-2): Research Analyst registration

- NISM Series XV exam, SEBI RA application
- Launch as algo strategy/signal platform
- Partner with broker for execution
- Cost: Rs 5-10L | Revenue in: 2-3 months

Phase 2 (Month 2-6): Authorized Person licenses

- Add AP licenses with 2-3 brokers
- White-label platform under broker licenses
- Cost: Rs 2-5L per broker

Phase 3 (Month 6-18): Full broker license (optional)

- Apply for SEBI stockbroker license
- Direct client acquisition
- Cost: Rs 50L+ initial, Rs 1Cr+ annually

5.3 Critical Regulatory Constraints

! **Algo trading (Apr 1, 2026):** All algos must be broker-hosted. No standalone APIs allowed

! **Order rate limit:** 10 orders/sec/exchange/client. Above = mandatory algo registration

! **Weekly expiry restricted:** Only 1 benchmark index per exchange (since Nov 2024)

! **F&O lot sizes increased:** Contract value Rs 15-20L (reduces retail speculation)

! **KYC/AML mandatory:** UBO identification, 5-year audit logs

! **Client fund segregation:** Escrow accounts mandatory

! **2FA + OAuth:** Required for all API access

6. TradePilot: The Opportunity

6.1 Positioning

"The platform that makes Indian traders profitable."

Not another order terminal. A profit-making machine with:

- AI trade scoring before every trade
- Risk guardrails that prevent catastrophic losses
- Backtesting so you validate before you risk real money
- Unified tools (no more 5 separate apps)
- Community with verified, accountable signal providers

6.2 Five Pillars of Differentiation

STRATEGY MAP – FIVE PILLARS



6.3 Target Market (Initial)

Segment	Size	Why They Switch
F&O traders losing money	10.5M	Platform that helps them STOP losing
Options traders using Sensibull + broker	3-5M	Unified experience, no tab-switching
Algo-curious retail traders	2-3M	No-code strategy builder + backtesting
Beginner traders (Gen Z)	5-10M/year	Education + guardrails + micro-investing
Profitable traders cobbling tools	500K-1M	All-in-one premium platform

6.4 Revenue Model

Stream	Price	Margin	Priority
Freemium (basic trading + education)	Rs 0	—	Acquisition
Pro subscription (AI scoring + backtesting)	Rs 499/mo	70%	Primary
Premium (algo builder + automation)	Rs 999/mo	75%	Growth
Brokerage (via broker partnership)	0.03-0.05%	15-30%	Volume
Data feeds (institutional-grade to retail)	Rs 199/mo	60%	Add-on
Certified courses	Rs 2K-5K	80%	Education

6.5 Competitive Moat

1. **AI models trained on Indian market data** — FII/DII flows, sector rotation, event impact
2. **Risk-first architecture** — no other Indian platform puts loss prevention at the core
3. **Unified workflow** — replacing 5-6 separate tools with one
4. **Community with accountability** — verified track records, not anonymous Telegram tips
5. **Regulatory-native** — built for SEBI's algo framework from day one

7. What Makes TradePilot "Futuristic"

7.1 Features No Indian Platform Has

Feature	TradePilot	Zerodha	Groww	Dhan
AI trade probability scoring	Yes	No	No	No
Historical options backtesting	Yes	No	No	No
No-code strategy builder	Yes	No	No	No
Risk guardrails (max loss/day)	Yes	No	No	No
Liquidity heatmaps	Yes	No	No	No
Unified portfolio analytics	Yes	Basic	Basic	Basic
Tax automation (ITR export)	Yes	No	No	No
Verified community signals	Yes	No	No	No
Voice trading	Phase 2	No	No	No
Fractional investing	Phase 2	No	No	No

7.2 The "Profit Probability" Screen

Before every trade, the trader sees:

BUY NIFTY 24500 CE @ Rs 152Profit Probability: **67%** [=====>]

Risk/Reward Ratio: 1:2.3

Max Loss (your limit): **Rs 5,000**

Position Size (recommended): 2 lots

Similar trades (last 90d): **73% profitable**FII sentiment: **Bullish** (net +2,400 Cr today)Volatility: **HIGH** (VIX 18.2, +12% from avg)**PLACE TRADE****BACKTEST FIRST****SKIP***This is what "built for profit" looks like.*

Sources

Market & Demographics

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- StockBrokers.com: Best Trading Platforms Awards

Technology

- QuestDB vs TimescaleDB Benchmark Results (2026)
- LiquidityFinder: Best AI Platforms for Trading & Analytics (2026)
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Regulatory

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- SEBI Algo Trading Rules (effective Apr 1, 2026)
- SEBI F&O Regulations (Nov 2024)
- SEBI Research Analyst Registration Guide

Community

- Reddit: r/IndianStreetBets, r/IndianStockMarket
- FreePressJournal: Zerodha 88% Unable to Trade (outage report)
- AlgoTest: Best Options Backtesting Tools in India